

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA1612

COURSE CODE: Data Warehousing and Data Mining

This report examines the applicability of clustering algorithms to five industry problems. For each problem, the industry context, engineering concepts, proposed solution, tools, and analytical outcomes are presented. The solutions use K-Means, Hierarchical Clustering, and DBSCAN according to the characteristics of each problem.

Q1. Retail Customer Segmentation

Industry Problem: An online retail company has thousands of customers but uses the same marketing strategy for everyone. The company wants to identify groups of customers with similar purchasing behavior so that personalized offers and campaigns can be created.

Application of Engineering Concepts: Use customer-level features such as annual spending, number of purchases, average order value, purchase frequency, and recency. The data should be cleaned, missing values handled, and numerical variables standardized before clustering.

Solution Design: Apply K-Means clustering. Select a suitable value of K using the Elbow Method and verify the quality using the Silhouette Score. Assign each customer to a cluster and describe the characteristics of each group, such as high-value customers, regular customers, occasional buyers, and low-engagement customers.

Tools and Techniques: Python, Pandas, NumPy, Scikit-learn, Matplotlib/Seaborn, Jupyter Notebook or Google Colab.

Analysis: The clusters can support targeted advertising, loyalty programs, personalized discounts, and customer retention. A limitation is that K-Means is sensitive to outliers and works best when clusters are relatively compact.

Suggested Visualization: Use an Elbow curve for selecting K and a 2D scatter plot after PCA showing the customer clusters.

Q2. Telecommunication Customer Churn Analysis

Industry Problem: A telecom company wants to understand groups of customers based on service usage and billing behavior. The goal is to identify customer segments that may require different retention strategies.

Application of Engineering Concepts: Use features such as monthly charges, call duration, data usage, number of complaints, contract duration, and service subscriptions. Normalize numerical attributes and encode categorical variables appropriately.

Solution Design: Use Hierarchical Clustering to create a dendrogram and examine relationships among customer groups. Select a practical number of clusters by cutting the dendrogram at a suitable distance level. Profile each cluster and identify groups with high charges, low usage, frequent complaints, or short contracts.

Tools and Techniques: Python, Pandas, Scikit-learn, SciPy for hierarchical clustering, Matplotlib, and Jupyter/Colab.

Analysis: Hierarchical clustering is useful because the dendrogram provides an interpretable view of relationships among customer groups. The results can guide retention offers and service improvements. For very large datasets, computational cost may become a limitation.

Suggested Visualization: Create a dendrogram and a cluster-size bar chart. A PCA scatter plot can also show the final customer groups.

Q3. Traffic Accident Hotspot Detection

Industry Problem: A city transport department receives accident records containing latitude, longitude, time, and accident severity. It needs to identify geographical accident hotspots to improve road safety and allocate emergency resources.

Application of Engineering Concepts: Spatial coordinates are the primary features. The data should be checked for invalid locations and duplicate records. Because the problem is spatial and may contain isolated observations, a density-based engineering approach is suitable.

Solution Design: Apply DBSCAN to the latitude and longitude coordinates. Tune the eps and minimum-samples parameters using the geographical scale of the city. Treat points classified as noise as isolated incidents. Dense groups represent potential accident hotspots.

Tools and Techniques: Python, Pandas, NumPy, Scikit-learn DBSCAN, GeoPandas or Folium for mapping, and Matplotlib.

Analysis: DBSCAN is appropriate because it does not require the number of clusters in advance and can detect irregularly shaped hotspots while identifying noise. The results can help authorities prioritize road lighting, signals, speed control, patrolling, and emergency response.

Suggested Visualization: Display the clusters on a city map using different markers or a hotspot map. Also show the number of accidents per cluster.

Q4. Manufacturing Machine Fault Detection

Industry Problem: A manufacturing plant collects sensor readings from machines, including temperature, vibration, pressure, power consumption, and operating speed. Unexpected machine behavior may indicate faults or maintenance requirements.

Application of Engineering Concepts: Sensor data should be cleaned, scaled, and checked for abnormal readings. Multiple sensor variables can be combined to represent machine operating conditions. Unsupervised clustering is useful when labeled fault data is limited.

Solution Design: Apply DBSCAN to separate dense normal operating patterns from isolated observations. Alternatively, K-Means can be used to identify normal operating states when the expected number of states is known. Points outside dense patterns can be flagged for inspection.

Tools and Techniques: Python, Pandas, Scikit-learn, NumPy, Matplotlib, and optionally a dashboard tool such as Power BI.

Analysis: DBSCAN can identify unusual sensor combinations without requiring predefined fault labels. Maintenance teams can use the clusters to investigate machines showing repeated abnormal patterns. A limitation is that parameter selection strongly affects DBSCAN performance.

Suggested Visualization: Use a PCA scatter plot with clusters and highlighted noise points. A time-series chart of sensor readings can be used to investigate abnormal periods.

Q5. Healthcare Patient Grouping

Industry Problem: A hospital wants to understand patient groups based on measurable clinical and service-use characteristics so that resources can be planned more effectively. The aim is not to replace medical diagnosis but to identify broad patterns for operational and analytical purposes.

Application of Engineering Concepts: Use appropriate de-identified variables such as age group, visit frequency, length of stay, selected laboratory measurements, and healthcare utilization indicators. Sensitive data must be protected, and features should be selected with domain experts.

Solution Design: Use K-Means for an initial segmentation when variables are numeric and clusters are expected to be compact. Compare the result with Hierarchical Clustering to examine whether meaningful subgroups appear at different levels. Evaluate clusters using Silhouette Score and domain interpretation rather than relying only on a numerical metric.

Tools and Techniques: Python, Pandas, Scikit-learn, SciPy, Matplotlib, and optionally Power BI for reporting. Use secure, de-identified datasets.

Analysis: The clustering results may reveal groups with different patterns of service utilization or length of stay, helping with resource planning and workflow design. Results should be treated as analytical groupings, not medical diagnoses, and validated by healthcare professionals.

Suggested Visualization: Use a cluster-size chart, PCA scatter plot, and a heatmap of average feature values for each cluster.

Overall Comparative Analysis

Industry Problem	Recommended Algorithm	Main Reason	Useful Tool	Visualization
Retail customer segmentation	K-Means	Fast segmentation of behavior groups	Scikit-learn	Elbow curve + PCA scatter
Telecom customer grouping	Hierarchical	Shows nested customer relationships	SciPy/Scikit-learn	Dendrogram
Accident hotspots	DBSCAN	Finds dense irregular spatial regions	GeoPandas/Folium	Cluster map
Machine fault detection	DBSCAN	Detects abnormal/noisy observations	Scikit-learn	PCA + time series
Healthcare patient grouping	K-Means + Hierarchical	Segmentation plus relationship analysis	Scikit-learn/SciPy	PCA + heatmap

Conclusion

Clustering algorithms provide practical solutions for industry problems where labeled outcomes are unavailable or difficult to obtain. K-Means is effective for efficient segmentation, Hierarchical Clustering is useful for understanding nested relationships, and DBSCAN is especially valuable for irregular clusters and noisy data. The correct algorithm should be selected based on data size, feature type, expected cluster structure, and the presence of outliers. Engineering analysis should combine quantitative evaluation, visualization, domain knowledge, and practical business or operational requirements.

Recommended Evaluation Measures

- Elbow Method – helps select an appropriate K for K-Means.
- Silhouette Score – measures how well observations fit their assigned clusters.
- Cluster Size and Feature Profiles – help interpret the practical meaning of each cluster.
- Dendrogram Inspection – supports selection of cluster levels in hierarchical clustering.
- Domain Validation – confirms whether the identified groups are meaningful for the industry problem.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation